

Human Performance Modeling - 5

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Application of Fitts' Law in UI Design

Keyboard Layout Optimization

Qwerty Layout



Q	W	E	R	T	Y	U	I	O	P
A	S	D	F	G	H	J	K	L	
	Z	X	C	V	B	N	M		

Qwerty is inefficient for one finger typing.

Optimization Objective Function

- Fitts' Law (Fitts 1954):

$$MT_{ij} = a + b \log_2 \left(\frac{D_{ij}}{W} + 1 \right)$$

MT_{ij} : Movement Time from Key i to Key j

D_{ij} : Distance from Key i to Key j

W : Key Width

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- Average time of typing a letter:

$$t = a + b \sum_i^{26} \sum_j^{26} P_{ij} \log_2 \left(\frac{D_{ij}}{W} + 1 \right)$$

P_{ij} : Frequency of an ordered letter pair i, j

Problem: Assuming that there is a special language called L which has three letters A, B, C only, we are designing a keyboard layout for this language. The digraph frequency of these three letters are shown in the following table. For example, the frequency of digraph BA is 0.3, and the frequency of digraph AB is 0.1. The Fitts' law parameters, which were learned from a previous study, are $a = 200$ ms, $b = 100$ (ms/bit). Please use the smaller-of 2D Fitts' law for calculation

2 nd letter 1 st letter	A	B	C
A	0.1	0.1	0.1
B	0.3	0.1	0.1
C	0.04	0.06	0.1

The keyboard layout is shown as below. Each key is a square. A user types with a pen on the soft keyboard.

A	C	B
---	---	---

1. What is the estimated movement time for typing the following digraphs, which is defined as the duration between the movement the pen leaves the first letter and moment the pen lands on the second letter? (hints: using smaller-of 2D Fitts' law)

AB, AC, BC, AA

2. What is the estimated average time of typing a letter on this keyboard layout?

Answers

1. duration for entering a digraph:

$$\text{AB: } 200 + 100 \log_2(3) = 358 \text{ ms}$$

$$\text{AC: } 200 + 100 \log_2(2) = 300 \text{ ms}$$

$$\text{BC: } 200 + 100 \log_2(2) = 300 \text{ ms}$$

$$\text{AA: } 200 \text{ ms}$$

2. What is the estimated average time of typing a letter on this keyboard layout?

$$0.1 \times 200 + 0.1 \times 358 + 0.1 \times 300 + 0.3 \times 358 + 0.1 \times 200 + 0.1 \times 300 + 0.04 \times 300 + 0.06 \times 300 + 0.1 \times 200 = 293.2 \text{ ms}$$

Searching for an optimized layout

This approach consists of numerous iterations. In each iteration, the procedure picks up two random keys on the keyboard and swaps their positions to reach a new configuration. The input speed of a new configuration is then estimated based on the Fitts-digraph model.

If the estimated input speed of the new layout is higher, the new layout will be kept. If not, there is still a chance to keep the new layout.

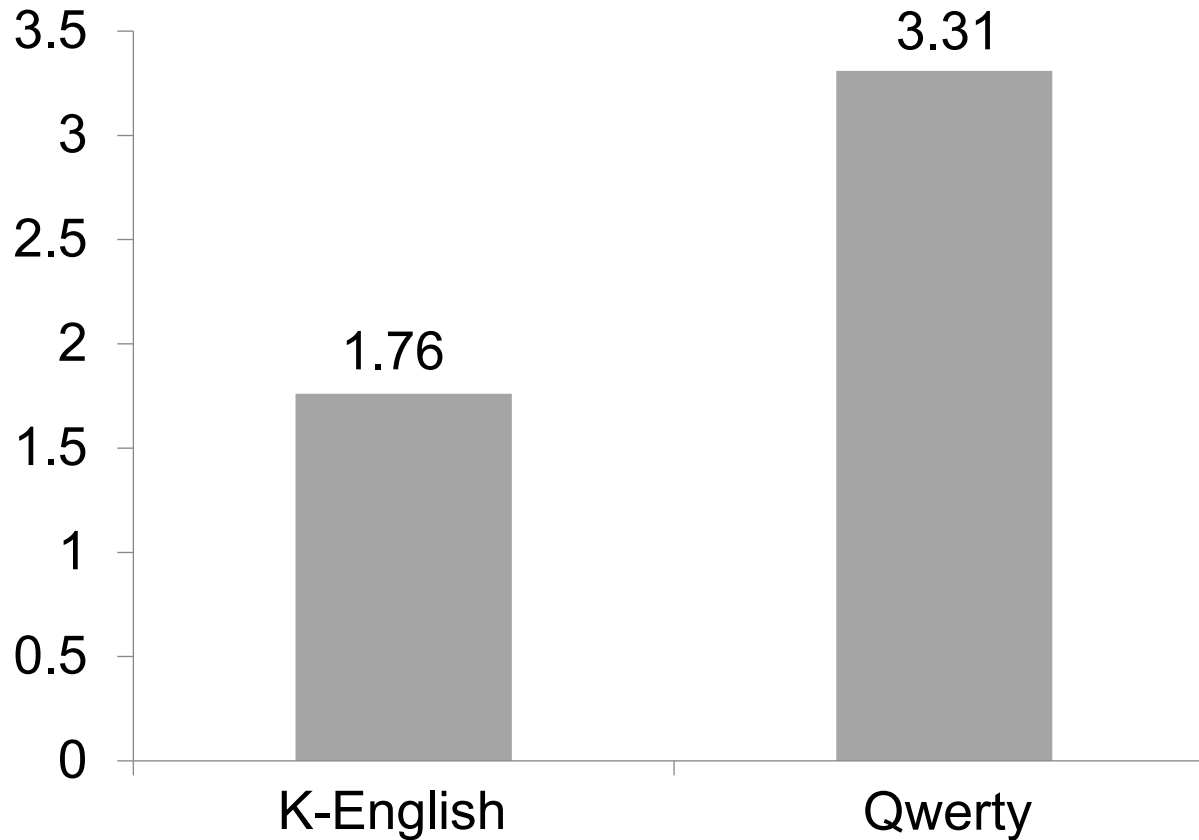
Layout Optimized for English

K-English

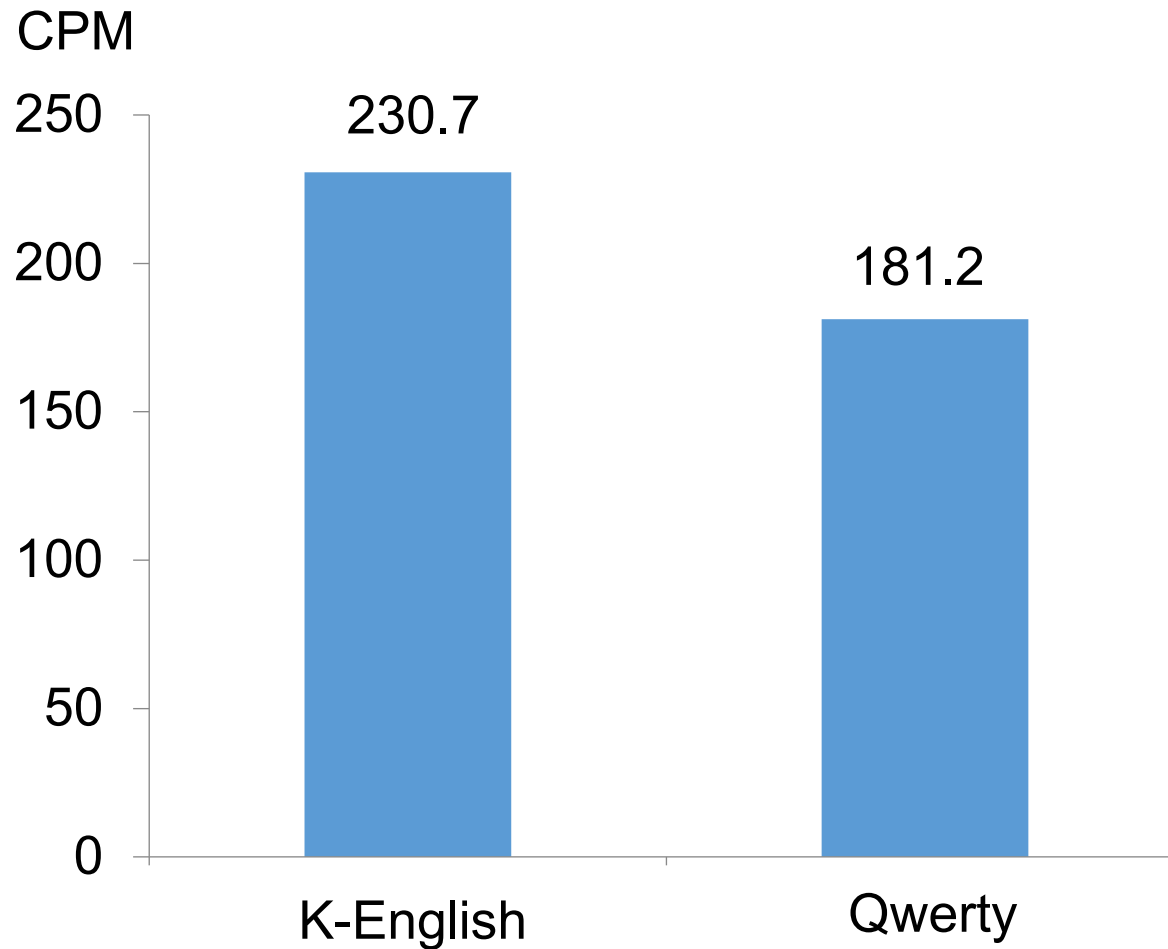
Z	J	D	G	K	
Y	L	N	I	C	
F	O	A	T	H	W
B	U	R	E	S	
Q	P	M	V	X	

Average Finger Travel Distance

Key Width



Typing Speed



Keyboard Layout Learning

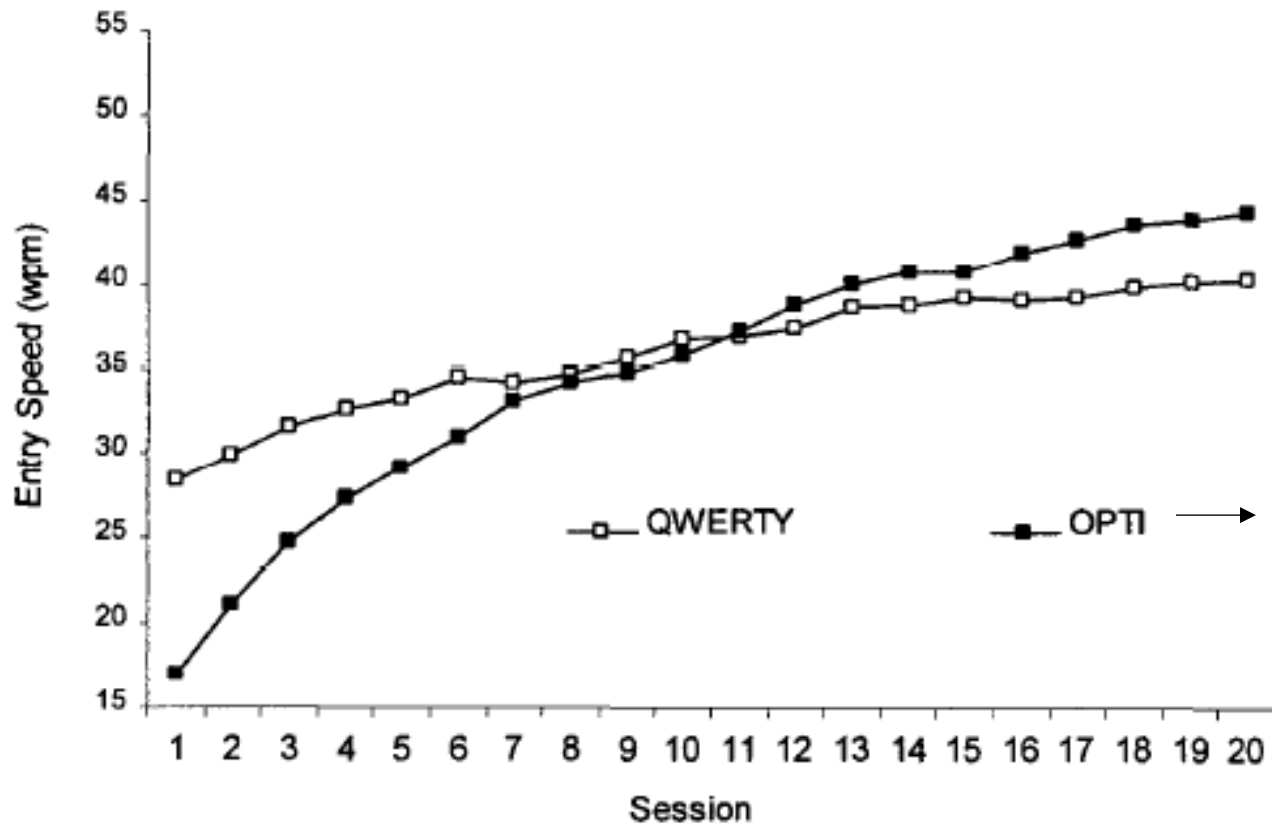


Figure 8. Entry speed by keyboard layout and session

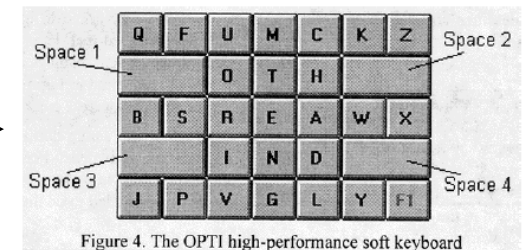
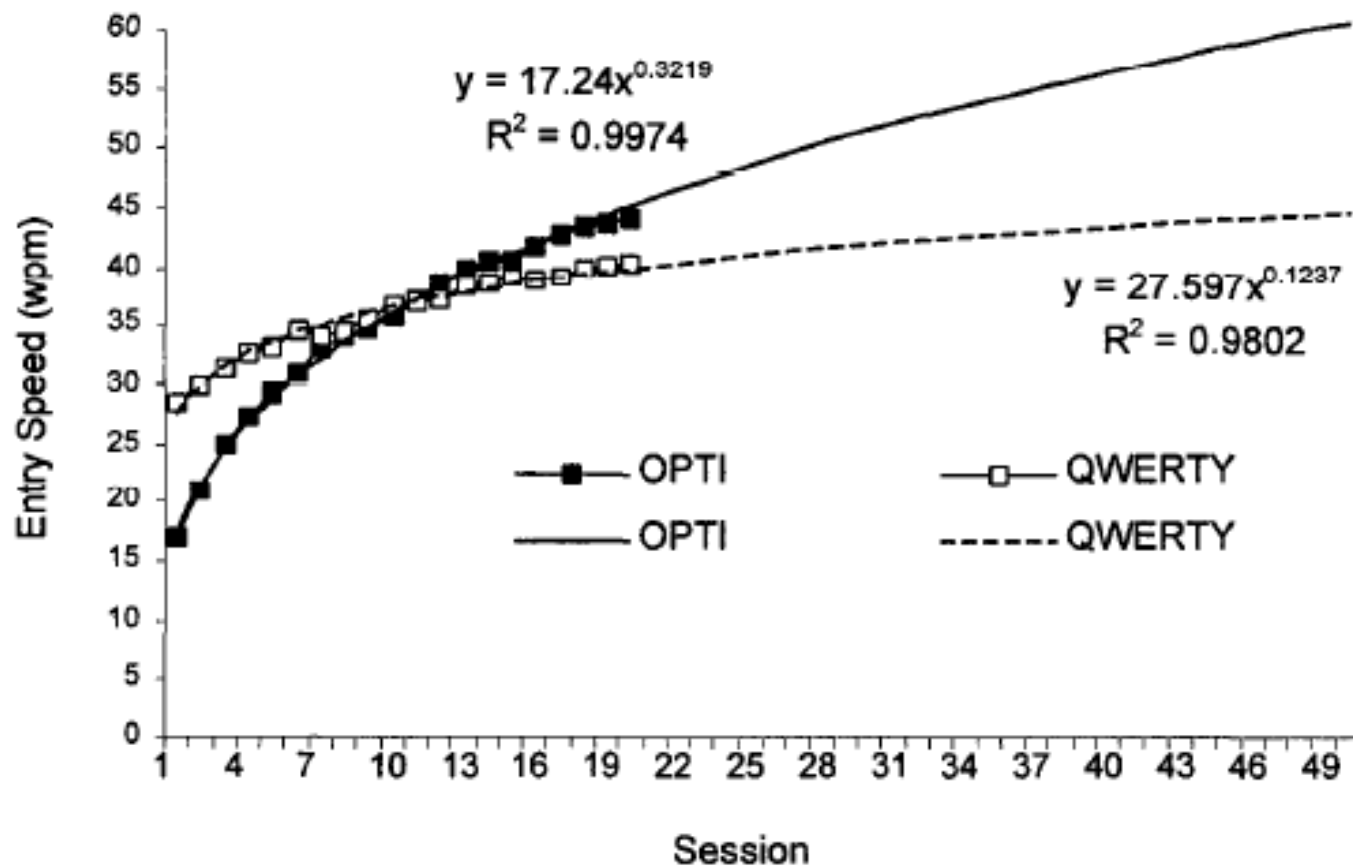


Figure 4. The OPTI high-performance soft keyboard

Keyboard Layout Learning



Keyboard Layout Learning

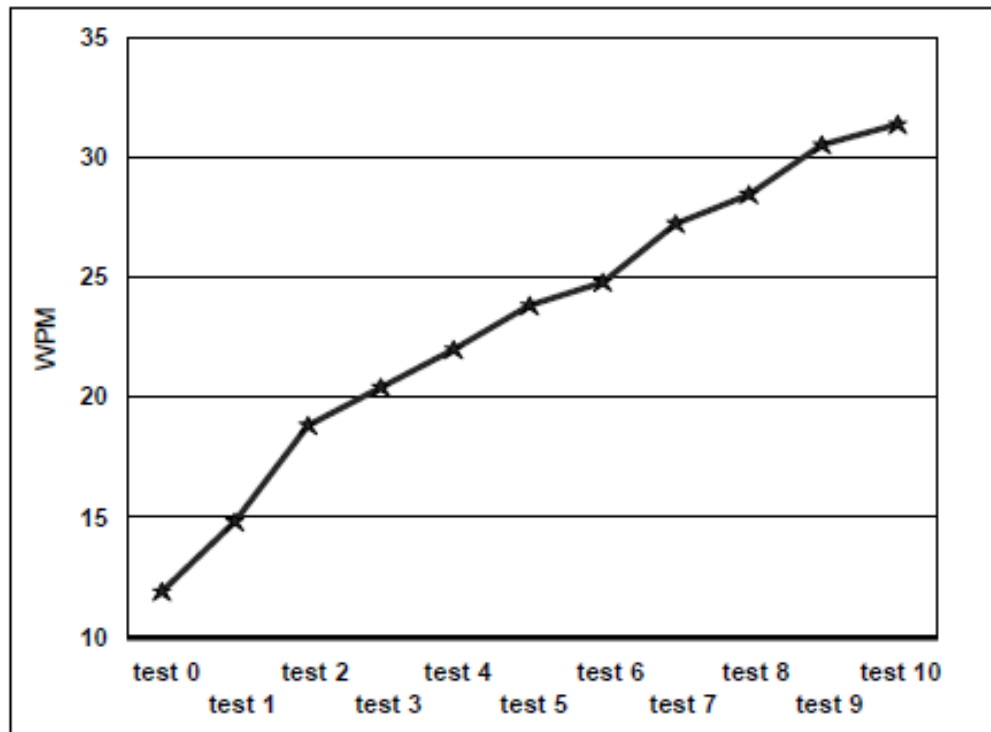
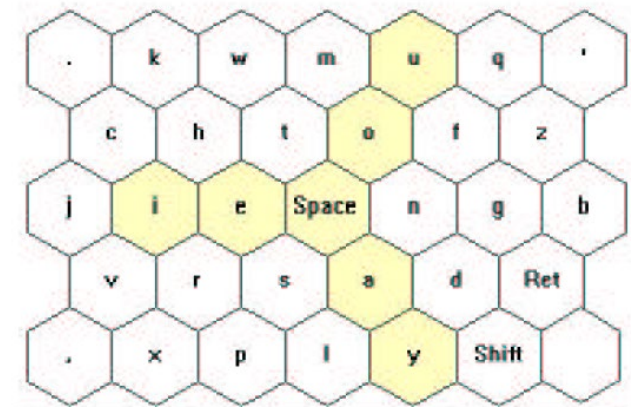


Figure 10. Learning curve in the experiment



$$\text{Speed} = 11.59 (N)^{0.41} \text{ (wpm)}$$

Zhai, Sue, and Accot. *Movement model, hits distribution and learning in virtual keyboarding*. ACM CHI2002

Quasi-Qwerty Layout

Q	W	D	R	T	U	Y	L	K	P
Z	A	S	E	H	N	I	O	M	
	X	F	V	C	G	B	J		

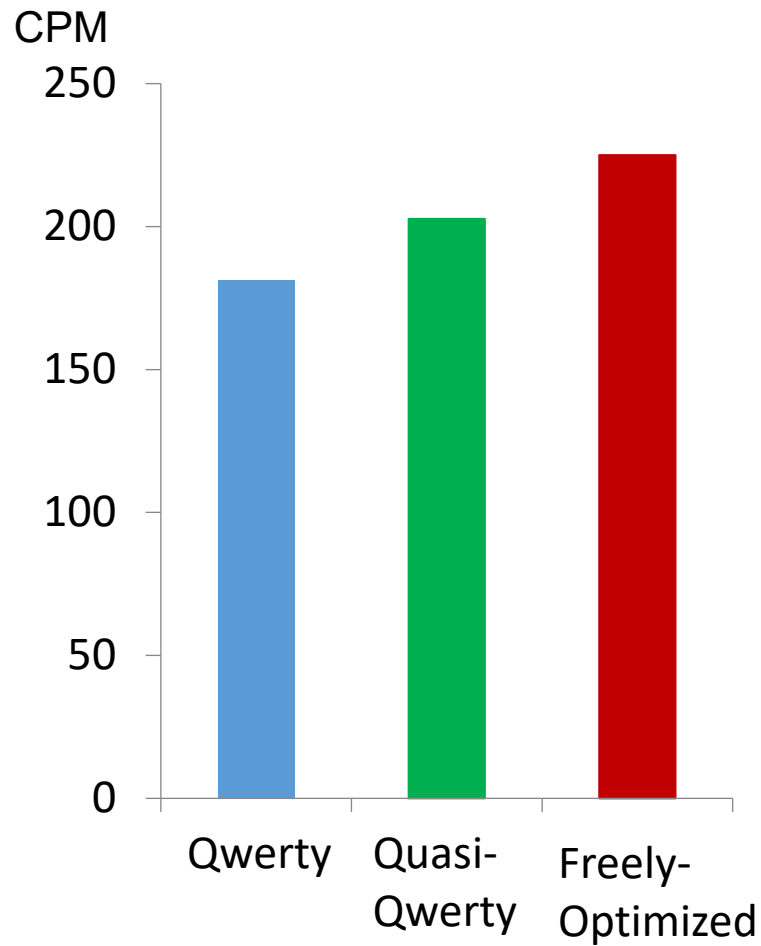
Quasi-Qwerty Layout

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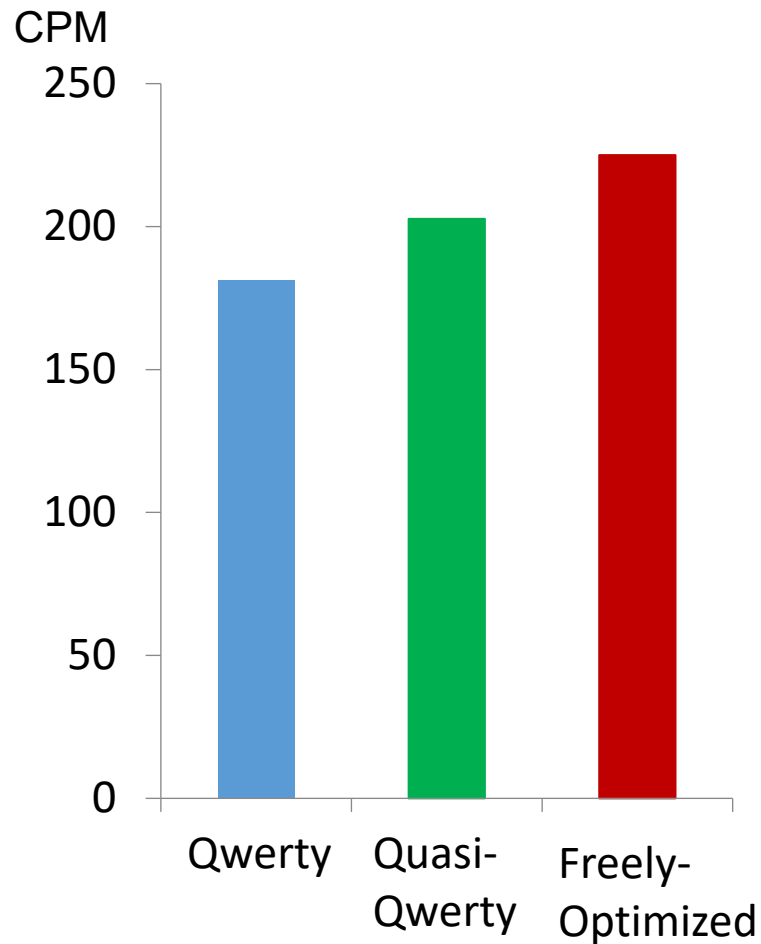
Qwerty

Q	W	E	R	T	Y	U	I	O	P
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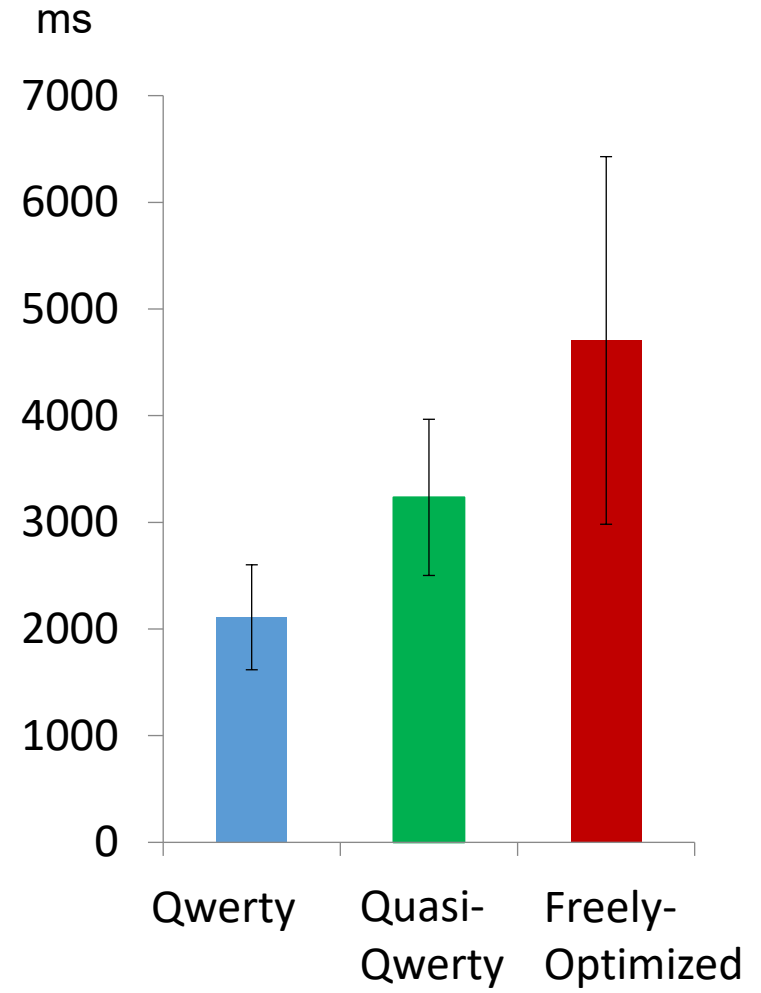
Expert Typing Speed



Expert Typing Speed



Initial Text Entry Time



Outline

- Crossing Law
- Steering Law

Crossing Law

- Not just pointing, how about crossing a boundary?



Johnny Accot and Shumin Zhai. 2002. More than dotting the i's --- foundations for crossing-based interfaces. CHI '02. ACM, New York, NY, USA, 73-80.

More than dotting the i's – Foundations for crossing-based interfaces

- Compare with pointing, Accot and Zhai propose 6 test conditions
 - 2 pointing tasks
 - 4 crossing tasks

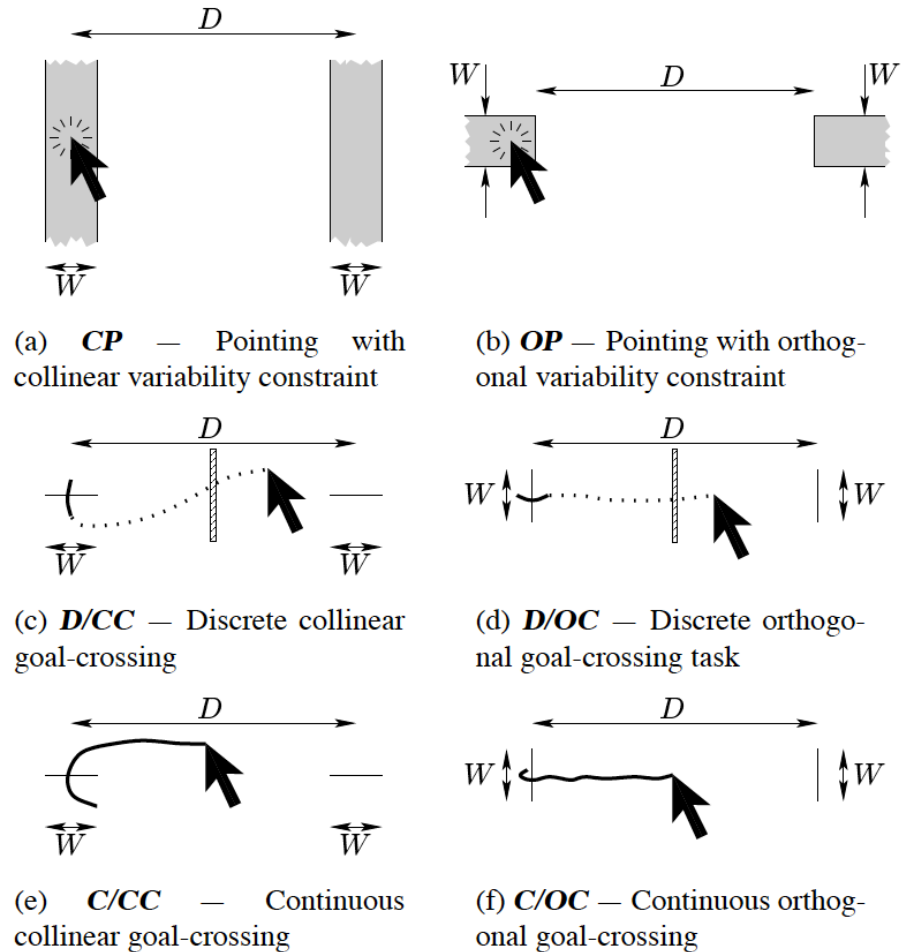


Figure 4: The six tested conditions. All tasks were reciprocal.

More than dotting the i's – Foundations for crossing-based interfaces

- Experiment result

$$CP: \quad T = 103 + 172 \times ID \quad r^2 = 0.998 \quad (4)$$

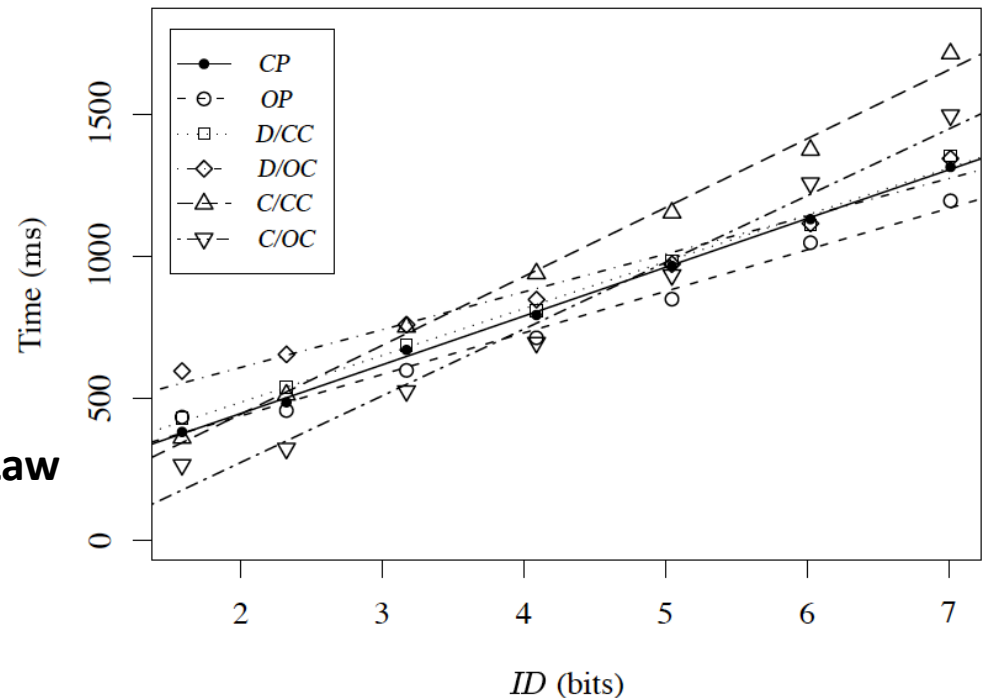
$$OP: \quad T = 145 + 146 \times ID \quad r^2 = 0.986 \quad (5)$$

$$D/CC: \quad T = 155 + 165 \times ID \quad r^2 = 0.994 \quad (6)$$

$$D/OC: \quad T = 342 + 133 \times ID \quad r^2 = 0.975 \quad (7)$$

$$C/CC: \quad T = -41 + 242 \times ID \quad r^2 = 0.995 \quad (8)$$

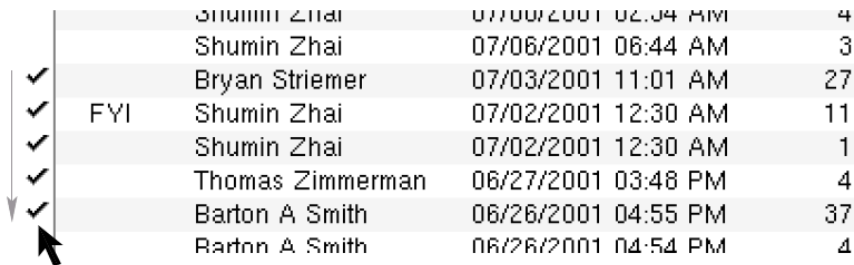
$$C/OC: \quad T = -196 + 235 \times ID \quad r^2 = 0.984 \quad (9)$$



Conclusion:
Crossing tasks can be modeled by Fitts' Law

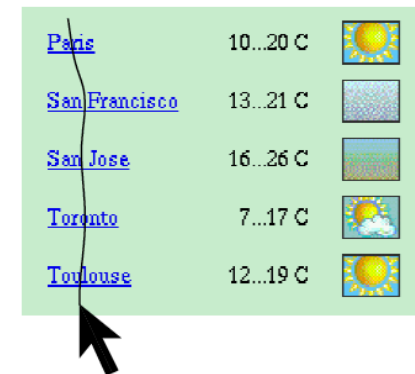
More than dotting the i's – Foundations for crossing-based interfaces

- Application in crossing interface



	Shumin Zhai	07/06/2001 02:34 AM	4
	Shumin Zhai	07/06/2001 06:44 AM	3
✓	Bryan Striemer	07/03/2001 11:01 AM	27
✓	FYI Shumin Zhai	07/02/2001 12:30 AM	11
✓	Shumin Zhai	07/02/2001 12:30 AM	1
✓	Thomas Zimmerman	06/27/2001 03:48 PM	4
✓	Barton A Smith	06/26/2001 04:55 PM	37
✓	Barton A Smith	06/26/2001 04:54 PM	4

Figure 13: Selection of multiple messages by a continuous goal-crossing action in Lotus Notes



Paris	10...20 C	
San Francisco	13...21 C	
San Jose	16...26 C	
Toronto	7...17 C	
Toulouse	12...19 C	

Figure 16: Crossing multiple city names to get their detailed weather forecast

Functionality

Point-and-Click

Crossing

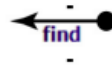
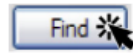
Functionality

Point-and-Click

Crossing

Buttons

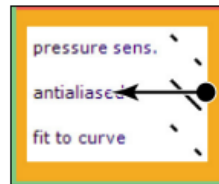
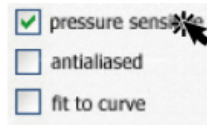
Standard



Radio

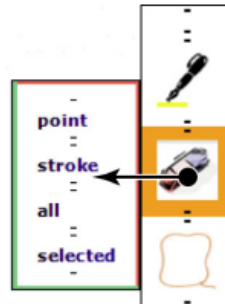
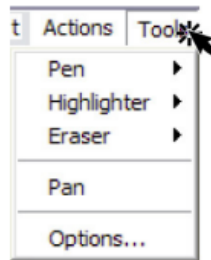


Checkbox



Menu

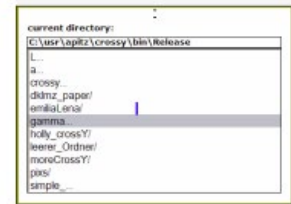
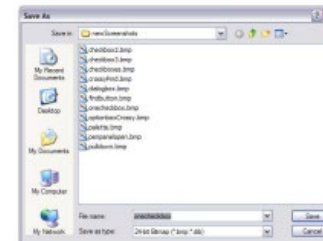
Pull-down



Scrollbar

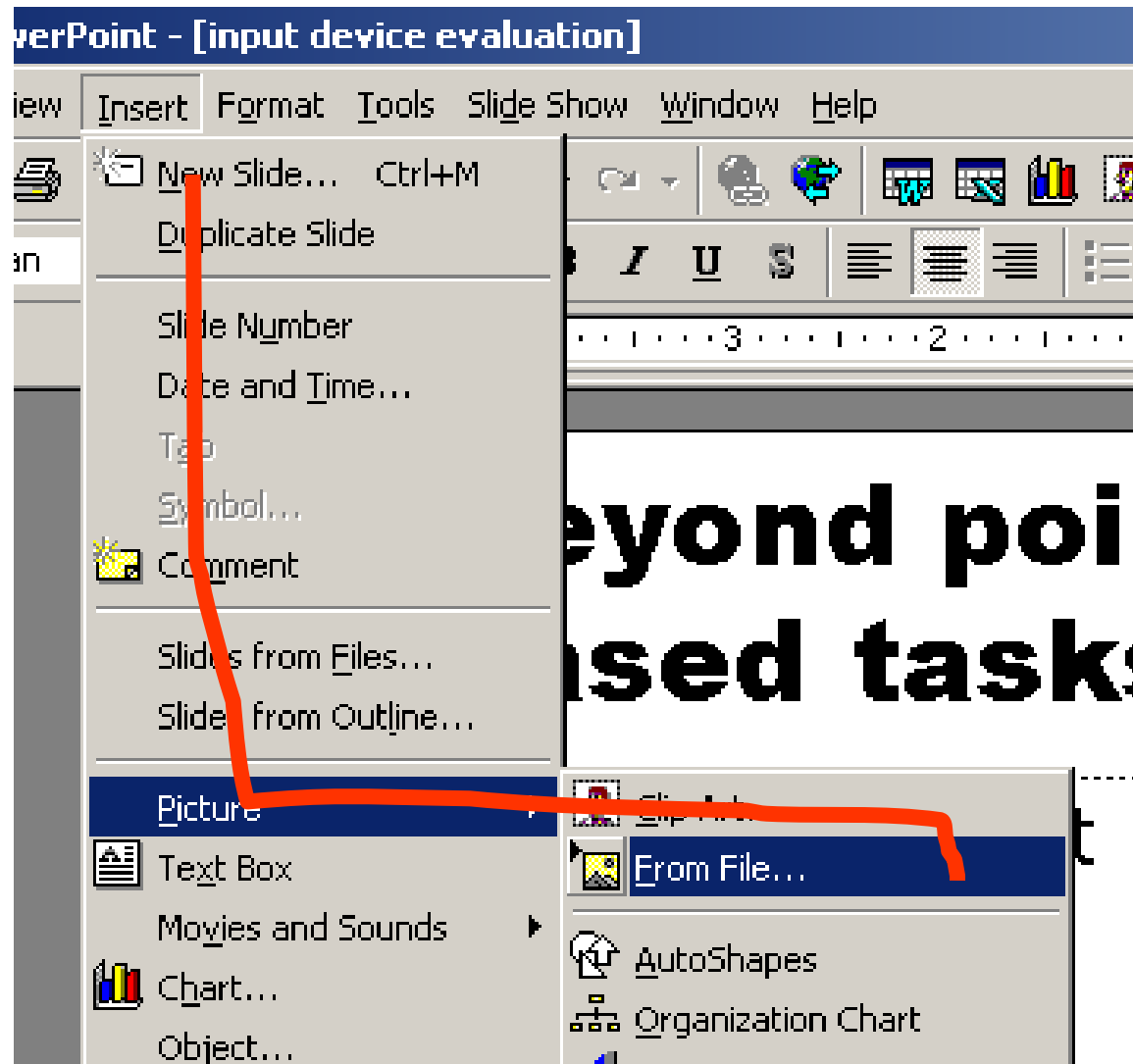


Dialog



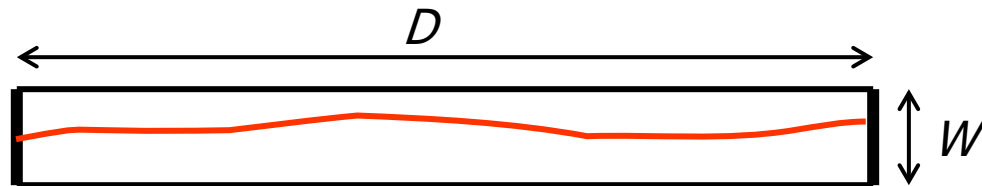
Georg Apitz and François Guimbretière. 2004. CrossY: a crossing-based drawing application. In Proceedings of the 17th annual ACM symposium on User interface software and technology (UIST '04). Association for Computing Machinery, New York, NY, USA, 3–12.

Beyond Pointing: Trajectory-based Tasks



Steering Law (a.k.a Accot-Zhai Law)

Johnny Accot and Shumin Zhai. 1997. Beyond Fitts' law: models for trajectory-based HCI tasks. In *CHI '97*. ACM, New York, NY, USA, 295-302.



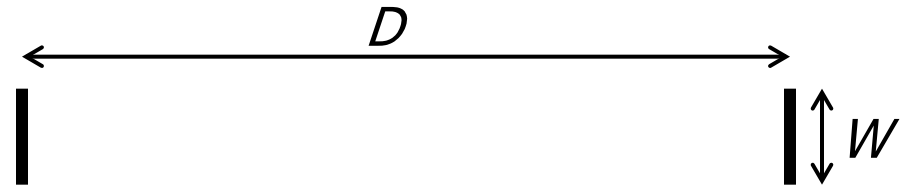
$$MT = a + b \frac{D}{W}$$

$$ID = \frac{D}{W}, \quad MT = a + bID,$$

From Targets to Tunnels...

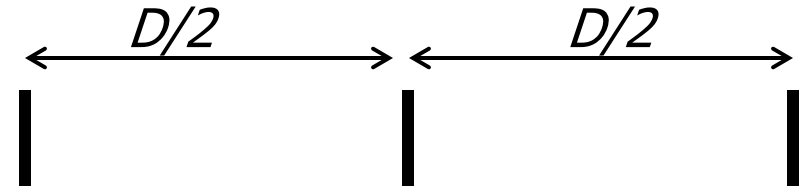
- 1 goal to pass through:

$$ID = \log_2\left(\frac{D}{W} + 1\right)$$



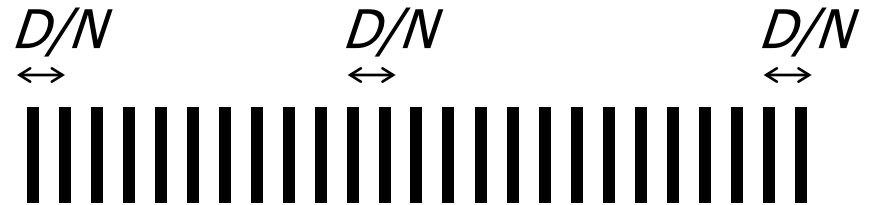
- 2 goals to pass through:

$$ID = 2\log_2\left(\frac{D}{2W} + 1\right)$$



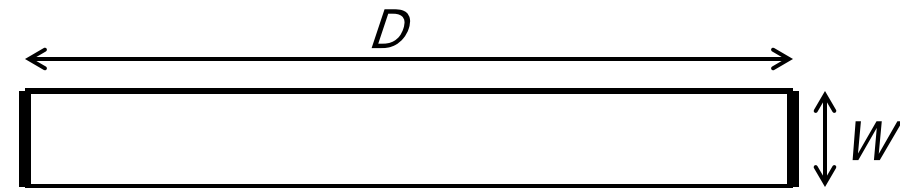
- N goals to pass through:

$$ID = N\log_2\left(\frac{D}{NW} + 1\right)$$



- ∞ goals to pass through:

$$ID_{\infty} = \frac{D}{W \ln 2}$$



Beyond Fitts' Law: Models for Trajectory-Based HCI Tasks

- The N^{th} step divides the amplitude A into N identical amplitudes $\frac{A}{N}$, as shown in Figure 4c. The difficulty to move from goal 1 to goal $N+1$ via goals 2, 3, ..., N is:

$$ID_N = N \log_2 \left(\frac{A}{NW} + 1 \right) \quad (5)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots,$$

Increasing con-
f is, the more
ugh all goals.
nel traveling"

task. The tunnel has length of A and width of W . (Figure 4d). It is also possible to determine the index of difficulty for the limit task by determining the limit of the index of difficulty recursion ID_N . Indeed, using a first order Taylor series expansion of $\log_2(1+x)$, we obtain:

$$ID_\infty = \lim_{N \rightarrow \infty} ID_N = \frac{A}{W \ln 2} \quad (6)$$

Therefore, such an analysis predicts that the difficulty to achieve this tunnel traveling task is not related to the logarithm of $\frac{A}{W}$ but to $\frac{A}{W}$. This leads to equation 7:

$$MT = a + b \frac{A}{W} \quad (7)$$

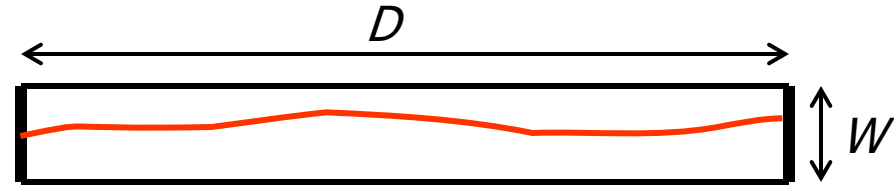
Steering Law (*Accot and Zhai, 1997*)

“Beyond Fitts’ Law: Models for trajectory based HCI tasks.”

Proceedings of ACM CHI 1997 Conference

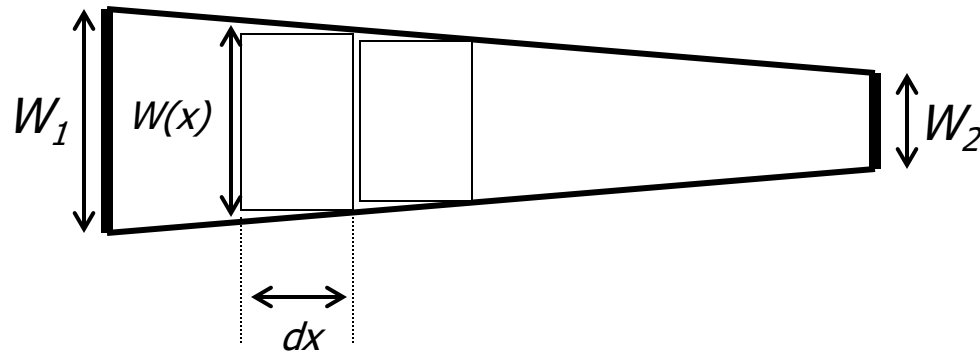
fixed width tunnel:

$$ID = \frac{D}{W}, \quad MT = a + b \frac{D}{W}$$



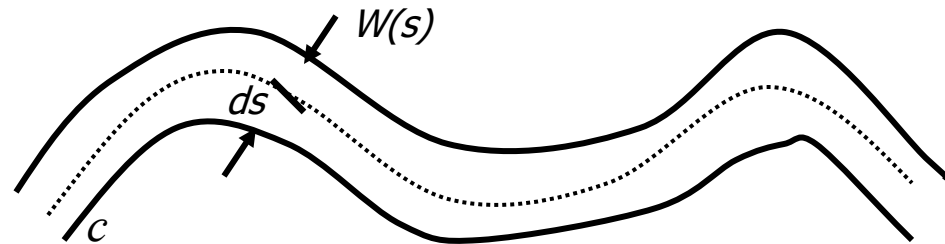
narrowing tunnel:

$$ID = \int_0^D \frac{dx}{W(x)}$$



general Steering Law:

$$ID = \int_c \frac{ds}{W(s)}$$



Some Results (from Accot, 1997)

